

x_1, x_2, x_3, x_4 features

x_1	x_2	y
2	1	+1
1	2	-1
-1	-1	-1
-2	-1	-1
1	-1	1

$$y = w_1 x_1 + w_2 x_2 + b$$

$$w_1 = 1, w_2 = 2, b = 0.5$$

$$y = w_1(2) + w_2(1) + 0.5$$

$$= 1(2) + 2(1) + 0.5$$

$$= 2 + 2 + 0.5$$

$$y = 4.5 = \frac{1}{1 + e^{-4.5}}$$

$$\text{tanh}(y) \rightarrow 0$$

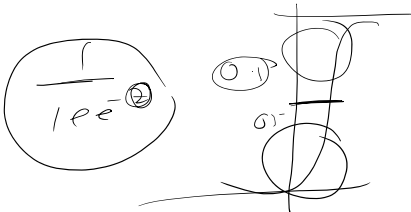
④

$$\textcircled{2} = w_1 x + w_0$$

linear ② for logit

$$y = \textcircled{2} \geq 0.5$$

$$\leq 0.5 \textcircled{1}$$



0, 1

linear regression

$$\text{output} \in (-\infty, \infty)$$

logit regression

$$\text{output} \in (0, 1)$$

$$y = w_1 x + w_0 \quad -\infty, \infty$$

$$p(\text{class} = 1)$$

$$p(y=1|x) = \frac{w_1 x + w_0}{1 + e^{-(w_1 x + w_0)}}$$

m,n	max	var
20	25	max
21	23	mm
25	26	man

$$p = p(y=1|x) \quad p \in (0, 1)$$

$$\text{odds} = \frac{p}{1-p} = \frac{\text{probability of success}}{\text{probability of failure}}$$

$$0.01 \quad \text{odds} = \frac{0.01}{1-0.01} = \frac{0.01}{0.99} = \frac{p}{1-p} = 0.01$$

$$0.75 = \frac{0.75}{1-0.75} = \frac{0.75}{0.25} = \textcircled{3} = \frac{1}{1-\phi}$$

1, 0, ..., 0, 0, ..., 0

$$2^0 = 1$$

$$2^1 - 1 = 2^1 - 2^0$$

$$= 2^1 - 2^0$$

$$\log\left(\frac{p}{1-p}\right) = \frac{(w_1 x + w_0)}{z}$$

$$\frac{0.9}{0.1}$$

4

$$= 2 \times 2$$

$$= \frac{4}{2} = 2$$

$$\log\left(\frac{p}{1-p}\right) = z$$

$$\frac{p}{1-p} = e^z$$

$$p = e^z(1-p)$$

$$p = e^z - p \cdot e^z$$

$$p + p \cdot e^z = e^z$$

$$p(1 + e^z) = e^z$$

$$p = \frac{e^z / e^z}{1 + e^z / e^z}$$

$$p = \frac{1}{\frac{1}{e^z} + 1} = \frac{1}{e^{-z} + 1}$$

$$p = \frac{1}{1 + e^{-z}}$$

Sigmoid function

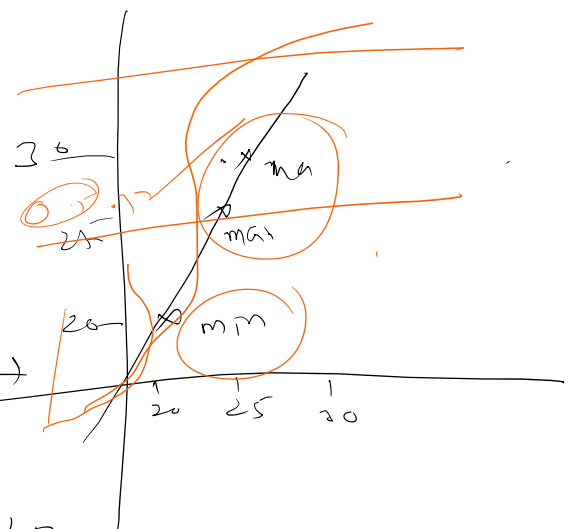
$$\min \quad \max$$

$$21 \quad 25$$

$$\frac{1}{1 + e^{-z}}$$

min	max	reorder label
20	24	
21	25	
24	26	
25		

$$\frac{\min}{\max}$$



$$\text{max temp} = w_1 \cdot \text{mintemp} + w_0$$

$$= (w_1)(20) + w_0 = 2$$

$$L = \frac{1}{N} \sum (y_{actual} - y_{predicted})^2$$

$N = \dots$ / order /

max temp $2(0.42)(25) + 15 = \dots$

label
max $\rightarrow 1$

min $\rightarrow 0$	max $\rightarrow 1$
2 0	0
2 1	1
2 4	1

$(-\infty, \infty)$

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$



$$p \geq 0.5 \text{ max} \rightarrow 1$$

$$p < 0.5 \text{ min} \rightarrow 0$$

linear classifier = $\frac{w^T \cdot x}{\|w\|} \geq 0$ / ≤ 0 ②

logistic regression $\rightarrow$